Zipf's Law for US Cities, 2000–2024

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Question

Zipf's law for cities says that the number of cities larger than S is proportional to $S^{-\zeta}$ with $\zeta = 1$: the second-largest city is half the size of the largest, the third a third, and so on. Gabaix (1999) reported ζ close to 1 for the 135 largest US metropolitan areas in 1991. The exponent matters beyond curve-fitting. An exponent of exactly 1 is what cities produce if they grow at rates unrelated to their size, so the law is a test of how cities grow, and $\zeta = 1$ is used to calibrate models of urban systems and to extrapolate the size of cities. Does the law hold in data published since 1999, for 2000 to 2024?

Result

It depends on how a city is defined, and the two common definitions fail in different ways. For metropolitan areas, the estimated exponent for the largest 135 is about 1.1 and is not statistically different from 1, but the rank-size relation is curved, so a single power law is rejected and the estimate moves from 1.42 to 0.90 as the cut-off widens. For incorporated cities (city proper), the rank-size relation is straight, but the exponent is about 1.45 and is statistically above 1. Both patterns are stable from 2000 to 2024. An exponent of 1, often treated as a constant of urban economics, therefore appears in these data only for metropolitan areas at particular cut-offs, and neither definition matches the simplest benchmark in Gabaix (1999), in which cities grow at rates unrelated to their size and the exponent is exactly 1. Figure 1 shows the rank-size plots for 2024 (panel a) and how the estimated exponent changes with the number of units included (panel b).

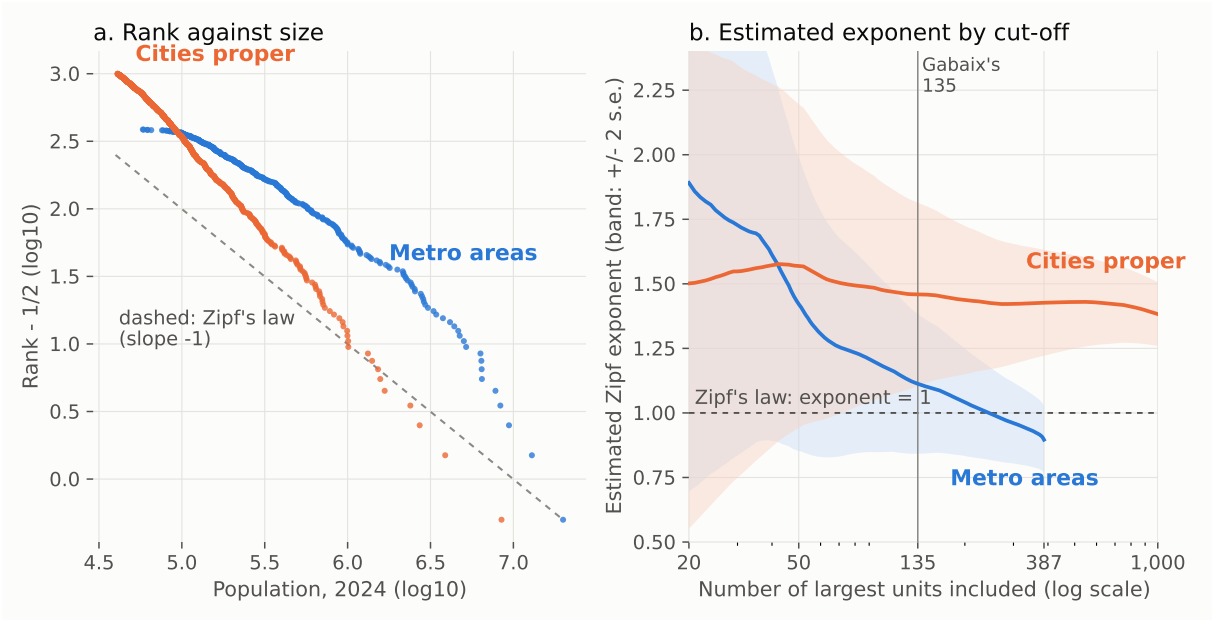


Figure 1: Zipf's law for US metropolitan areas and incorporated cities, 2024. Panel a: rank minus one half against population, both on log scales, for all 387 metropolitan areas and the largest 1,000 cities; the dashed line has slope -1, the Zipf benchmark. Panel b: estimated exponent ζ for the largest N units, with bands of ± 2 standard errors ($\zeta\sqrt{2/N}$); the dashed line is $\zeta = 1$ and the vertical line marks $N = 135$, the cut-off in Gabaix (1999).

Why it matters

Theory. Zipf's law is one of the most cited regularities in economics, and it is a target that models of urban growth are built to hit. Gabaix (1999) shows that if cities grow at rates independent of their size (Gibrat's law), their sizes converge to a power law with $\zeta = 1$. A curved rank-size relation for metropolitan areas means that size-independent growth does not describe them: something, such as congestion at the top or entry of new metros at the bottom, makes growth depend on size. Models calibrated to match "$\zeta = 1$" are matching a number that the data deliver only at particular cut-offs.

Measurement. The answer depends on the unit. Metropolitan areas approximate labour markets; cities proper are administrative units that raise taxes and provide services. A planner or modeller who borrows "$\zeta = 1$" for municipalities would overstate how large the largest municipalities are relative to the rest, since an exponent of about 1.45 means sizes fall off more slowly with rank. Under Zipf's law the 100th-largest city would be one tenth the size of the 10th-largest; in 2024 Norfolk (100th) is 0.23 times the size of Jacksonville (10th).

Method. The same estimator, applied to the same country and year, gives exponents from 0.9 to about 1.9 depending on the unit and the cut-off (Figure 1, panel b). Claims that a power law "holds" for cities, firms or wealth are more informative when they report the cut-off, the standard error and a curvature test.

Data and design

Data. Census Bureau population estimates files, which contain census counts and annual estimates. Metropolitan areas are the 384 to 387 Metropolitan Statistical Areas in current delineations (micropolitan areas and metropolitan divisions are excluded), for the 2010 census, the 2020 census base and the July 2024 estimate. Cities are incorporated places (summary level 162) for the 2000 and 2020 census bases, the 2010 census and the July 2024 estimate. Data before 2000 were not used, so every year in the sample postdates Gabaix (1999).

Estimator. For the largest N units, ordered by population $S_{(1)} \geq \cdots \geq S_{(N)}$, the exponent is minus the slope from the least-squares regression of $\log(i - \frac{1}{2})$ on $\log S_{(i)}$, with standard error $\zeta\sqrt{2/N}$ (Gabaix and Ibragimov, 2011). The "t vs 1" column tests $\zeta = 1$.

Curvature test. Gabaix and Ibragimov (2011) add the term $c(\log S - \gamma)^2$, with γ chosen to make it orthogonal to $\log S$. Under a power law $c = 0$; the statistic $c/(\zeta^2\sqrt{1/(2N)})$ is approximately standard normal, and an absolute value above 1.95 rejects the power law at the 5% level.

		Largest 135				ζ at other cut-offs		
Unit	Year	ζ	s.e.	t vs 1	Curvature	Top 50	Top 250	Wide
Metro areas	2010	1.14	0.14	1.0	−2.3	1.42	1.00	0.92
	2020	1.11	0.14	0.8	−2.7	1.42	1.00	0.90
	2024	1.11	0.14	0.8	−2.7	1.42	1.00	0.90
Cities proper	2000	1.41	0.17	2.4	−0.3	1.52	1.37	1.34
	2010	1.45	0.18	2.5	−0.2	1.54	1.41	1.36
	2020	1.45	0.18	2.5	−0.3	1.56	1.42	1.38
	2024	1.46	0.18	2.6	−0.3	1.57	1.43	1.38

Table 1: Zipf exponents for US metropolitan areas and incorporated cities. "Wide" is all metropolitan areas (384 in 2010, 387 later; smallest about 55,000–58,000 people) or the largest 1,000 cities (smallest about 32,000–41,000). The curvature statistic is for the largest 135; values beyond ±1.95 reject a power law. For the largest 250 metropolitan areas it is −3.0 in every year.

For metropolitan areas, ζ for the largest 135 ranges from 1.11 to 1.14 with a standard error of 0.14, so the data do not reject $\zeta = 1$. The curvature statistic is −2.3 to −2.7, and −3.0 for the largest 250, so the log rank-size relation is concave: the very largest metros are fewer than a straight line implies (local ζ of 1.42 among the top 50) and mid-sized metros are more numerous (0.90 across all). The point estimate is within 0.05 of 1 only for cut-offs between about 190 and 320 metropolitan areas (Figure 1, panel b). The standard errors are wide enough that no single cut-off rejects $\zeta = 1$; the evidence against a single power law comes from the curvature test. A reported ζ near 1 is therefore partly a consequence of the cut-off. For cities proper, the curvature statistic is close to zero at every date, but ζ is 1.41 to 1.46 for the largest

135 (t of 2.4 to 2.6 against 1) and about 1.37 for the largest 1,000 (t near 6). Over 2000–2024 the city exponent rose slightly, meaning large cities became a little more similar in size, but the change is well within one standard error.

Caveats

Metropolitan areas exist only above 50,000 people in the urban core, which truncates the lower tail and contributes to the flattening seen in Figure 1. City-proper populations depend on municipal boundaries; New York City and Jacksonville, for example, have consolidated city and county governments. Across all 19,475 incorporated places the distribution is close to lognormal, with an estimated ζ near 0.5, so the city results apply only to the upper tail. The 2024 values are estimates rather than counts. The standard errors assume independent ranks and are approximate; the curvature test is asymptotic.

Relation to the literature

Each component has been reported before. Eeckhout (2004) shows that the full distribution of US places is lognormal. Soo (2005) rejects Zipf's law for a majority of 73 countries. Bee, Riccaboni and Schiavo (2013) find that the largest US cities look lognormal rather than Pareto. Rozenfeld, Rybski, Gabaix and Makse (2011) find ζ close to 1 when cities are built from population clusters rather than administrative units. This note updates the estimates to 2020 and 2024 and sets the two administrative definitions side by side.

References

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Data: US Census Bureau population estimates files (public). Drafted with AI assistance and checked before publication.